AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-1042*

Team Name: *IdeaForge*

Domain: *Domain 5:Food Security,Agriculture Rural Transformation*

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Presentation Outline

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Problem Statement

The Core Flaw:
The global supply chain manages living, biological produce using static, printed expiration dates.

The Reality:
Food decays dynamically based on environmental conditions, not calendar math.

The Impact:
Rigid First-In, First-Out (FIFO) logic contributes to food waste, lost revenue, and environmental strain

Problem Understanding

- Who experiences this problem?
  Affected Users: Farmers, retailers, distributors, logistics providers, and consumers face losses due to perishable food spoilage.

- What causes the problem?
  Root Causes: Temperature fluctuations, humidity, improper storage, transportation delays, and fixed expiry dates that cannot reflect changing freshness.

- How does it affect users or stakeholders?
  Consequences: Increased food waste, financial losses, inefficient inventory management, and unnecessary environmental impact.

Existing Gaps & Challenges

- Existing gap or limitation 1
  Static Expiry Dates: Printed dates cannot reflect real-time changes in food freshness.

- Existing gap or limitation 2
  Rigid FIFO Logic: Products are prioritized by arrival order rather than actual decay risk.

- Main challenge that remains unsolved
  Lack of Dynamic Monitoring: Changing temperature, humidity, and delivery delays are not adequately reflected in inventory decisions.

Our Proposed Solution

- Core idea or product
  Core Idea: FreshPulse AI is an intelligent logistics engine that dynamically evaluates the freshness and decay risk of perishable products.

- How the solution works at a high level
  How It Works: It combines product freshness, environmental conditions, expiry information, local demand, and delivery urgency to prioritize stock and recommend suitable distribution decisions.

- Main value delivered to users
  Main Value: Reduces avoidable food waste, minimizes financial losses, and improves the efficiency of perishable-food distribution.

FreshPulse AI: System Architecture

Data Collection

Temperature, Humidity,

Dashboard and Alerts

Product Sta-
tus, Warnings
and Recom-

Technical Approach: Technology Stack

Frontend: HTML, CSS, JavaScript and Streamlit for an interactive dashboard.

Backend: Python for data processing, freshness estimation and business logic.

Database: SQLite for the prototype; PostgreSQL for future deployment.

AI/ML: Pandas and NumPy for data analysis; Scikit-learn for predictive modelling and spoilage-risk classification.

Mapping and Logistics: OpenStreetMap with a routing service for nearby destinations and redistribution planning.

Hardware (Optional): Temperature and humidity sensors for real-time storage monitoring.

Expected Outcome: Dynamic freshness assessment, earlier spoilage warnings, reduced food waste and improved local redistribution.

Innovation & Uniqueness

- What is new or different?
  Dynamic Freshness Intelligence: Evaluates changing food decay risk instead of relying only on printed expiry dates.

- Why is this approach suited to the problem?
  Condition-Aware Decisions: Considers temperature, humidity, storage conditions, and transportation delays to prioritize products.

- What is the key differentiator?Hyper-Local Distress Logistics: Connects products nearing spoilage with nearby demand and suitable delivery options to reduce avoidable waste.

Feasibility & Scalability

- **Feasibility:** A prototype can be developed using Python, freshness and inventory data, weather or sensor inputs, and a simple dashboard.
- **Scalability:** The system can expand from one grocery store to multiple retailers, warehouses, and regional distribution networks.
- **Risks:** Inaccurate data and limited sensor availability may affect predictions. Data validation, regular monitoring, and pilot testing can help reduce these risks.

Cost for Product (Optional)

- Development or hardware cost: Development Cost: 0–2,000 using free tools and student-built software
- Hosting or maintenance cost: Hosting and Maintenance: 0–1,000 for a basic prototype using free-tier services.
- Estimated total: Estimated Prototype Total: 0–3,000, excluding existing laptop costs and any optional sensors

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Use Case 1 – Grocery Stores: Identify products with rising spoilage risk and prioritize them for quick sale or distribution.
- Use case 2: Use Case 2 – Warehouses: Monitor storage conditions and prioritize inventory based on freshness and estimated decay risk.
- Use case 3: Use Case 3 – Local Food Distribution: Connect nearby demand with perishable products approaching spoilage and recommend faster delivery options.

Impacts and Benefits

- [Benefit to the target users]
- [Social, economic, or environmental impact]
- [Expected measurable outcome]

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
1	2	Zero Hunger	Reduces food waste by directing perishable food to nearby demand before it spoils.
12	3	Responsible Consumption and Production	Minimizes food waste through dynamic freshness monitoring and smarter inventory management.
13	4	Climate Action	Reduces greenhouse gas emissions associated with

Individual Member Contribution

Member	Contribution or responsibility
Kanishk S	Problem Research
Tarun K S	Solution Design
SreeVarshini R	Presentation and Solution Design
Shredha V	Presentation and Research

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